

SIMATS ENGINEERING

ASSESSMENT TOOL – 1

Course: Cryptography and Network Security (CSA51)

Name: T.LAKSHMI PRASAD NAIDU

Reg. No: 192324187

1. Scenario: Event Communication Security

(a) Encrypt "MEET AT NOON" using Caesar Cipher (Shift = 4)

Shift every letter 4 positions forward.

Plaintext M E E T A T N O O N

Ciphertext Q I I X E X R S S R

Encrypted Message:

QIIX EX RSSR

(b) Decryption Process

An authorized member shifts every ciphertext letter **4 positions backward**.

Example:

- Q → M
- I → E
- X → T
- R → N
- S → O

Therefore,

QIIX EX RSSR → MEET AT NOON

2. Scenario: Intercepted Suspicious Message

Ciphertext:

GSRH RH Z HVXIVG

(a) Original Message

This is encrypted using the **Atbash Cipher**.

Cipher G S R H R H Z H V X I V G

Plain T H I S I S A S E C R E T

Original Message:

THIS IS A SECRET

(b) Justification

- Atbash cipher replaces every alphabet with its opposite letter.
- $A \leftrightarrow Z$
- $B \leftrightarrow Y$
- $C \leftrightarrow X$
- $D \leftrightarrow W$
- $E \leftrightarrow V$

The repeating pattern "RH" becomes "IS", confirming the message.

3. Scenario: Secure Messaging System

Plaintext:

HELLO

Keyword:

KEY

(a) Vigenère Cipher Encryption

Repeat the keyword:

Plaintext H E L L O

Keyword K E Y K E

Alphabet values:

Letter Value

H 7

E 4

L 11

L 11

O 14

Keyword values:

Letter Value

K 10

E 4

Y 24

K 10

E 4

Encryption:

- $H + K = R$
- $E + E = I$
- $L + Y = J$
- $L + K = V$
- $O + E = S$

Ciphertext:

RIJVS

(b) Role of Repeating Keyword

- The keyword repeats until it matches the message length.
- Each keyword letter determines the shift value.
- Different shifts make the cipher stronger than the Caesar cipher.

4. Scenario: Suspicious Digital Asset Investigation

(a) Detecting Hidden Data

Investigators can detect hidden data by:

1. Performing statistical analysis of pixel values.
2. Comparing file size and metadata.
3. Using steganalysis tools.
4. Checking Least Significant Bit (LSB) modifications.

(b) Two Security Improvements

1. Encrypt the secret message before embedding it.
2. Use randomized or adaptive LSB embedding to make detection difficult.

5. Scenario: Legacy Encryption in Corporate Communication

Ciphertext:

WECRLTEERDSOEFEAOCAIVDEN

(a) Original Message

This is a **Rail Fence Cipher (3 Rails)**.

After decryption,

WE ARE DISCOVERED FLEE AT ONCE

(b) Why is it Vulnerable?

- Uses only letter rearrangement.
- No secret substitution of letters.
- Easy to break using brute-force methods.
- Modern computers can decrypt it quickly.

(c) Practical Enhancement

Use **AES (Advanced Encryption Standard)** or encrypt the message with a modern symmetric encryption algorithm before transmission.

Conclusion

Classical encryption methods such as Caesar, Atbash, Vigenère, and Rail Fence are useful for learning the fundamentals of cryptography. However, they are vulnerable to cryptanalysis and brute-force attacks. Modern algorithms like AES and RSA provide significantly stronger confidentiality and are recommended for secure communication.